Persistent Adaptive Cognition

Investigating the Computational Role of Persistent Internal State in Adaptive Artificial Agents

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ABSTRACT

Artificial intelligence systems are increasingly capable of reasoning, planning, using tools, and acting over extended periods, yet an important architectural question remains open: how much does an agent gain from maintaining an internal state that persists continuously through time, rather than reconstructing its state from a bounded history of recent observations?

This paper develops a research program around that question. The motivating long-term hypothesis is that if consciousness is realizable in an artificial system, it may depend on a particular organization of computation rather than on scale alone an organization involving persistent internal state, recurrent causal dynamics, information integration, self-modeling, adaptive learning, and closed-loop interaction. The present work deliberately tests a narrower claim first.

We introduce Persistent Adaptive Cognition (PAC), a sequence of experiments designed to isolate the computational contribution of persistent internal state in partially observable, non-stationary environments. PAC-1 compares a stateless multilayer perceptron, a finite-context Transformer, and a recurrent GRU under hidden regime changes, noisy observations, and reward feedback. In the initial run, the Transformer achieved the highest mean reward (0.308), followed by the GRU (0.283) and stateless model (0.173). The Transformer also recovered from 526 detected regime changes with a mean adaptation time of 133.2 steps, whereas the current GRU evaluation registered no recovered changes under the chosen recovery criterion.

These results do not support the stronger claim that persistent recurrent state is superior to finite-context memory. They do, however, provide preliminary evidence that temporal information substantially improves performance over a stateless baseline and expose methodological questions that must be resolved before a stronger conclusion can be drawn.

The paper therefore treats PAC-1 as a pilot rather than a confirmation. It proposes a controlled follow-up program involving state-reset ablations, matched training conditions, multiple random seeds, longer temporal dependencies, delayed consequences, distractor information, and systematic scaling of finite context. The broader objective is not to establish machine consciousness through a single benchmark, but to build an empirical ladder from persistence and adaptation toward metacognition, self-modeling, global integration, agency, and only if evidence warrants it questions of artificial consciousness.

1. INTRODUCTION

The most interesting questions in artificial intelligence are increasingly moving away from whether a model can produce an intelligent answer and toward what kind of system can sustain intelligence over time.

A contemporary model can often perform an impressive piece of reasoning while it is being queried. It can retrieve information, write code, use tools, plan a sequence of actions, and in some settings maintain a large amount of

context. But an agent operating continuously in the world faces a different problem. It must decide what to remember, what to forget, how to update its beliefs when the environment changes, how to distinguish a change in the world from a change in its own uncertainty, and how to maintain useful internal variables across time.

This distinction matters because an agent's history is not the same thing as an internal state. A finite context window gives a system access to a record of recent observations. A recurrent system, by contrast, can maintain a compact state that is transformed at every timestep. The latter is not automatically better. A poorly trained recurrent system may forget useful information, become unstable, or learn a representation that is less useful than simply looking back at recent observations. The scientific question is therefore not whether recurrence sounds more biological, but whether persistent state creates measurable computational advantages under conditions where persistence should matter.

That question sits inside a larger research program. One motivation comes from computational approaches to mind, particularly the idea that mental functions may depend on the causal organization of a system rather than on the biological material in which that organization is implemented. Chalmers' work on computational implementation, for example, frames computation in terms of a correspondence between causal organization in a physical system and the formal structure of a computation. This provides a useful conceptual bridge between cognitive theory and engineered systems, while leaving open difficult questions about which computational organizations are sufficient for consciousness.

For engineering purposes, the important implication is methodological. Instead of beginning with the claim that a machine is or is not conscious, one can ask whether particular computational organizations produce capabilities associated with increasingly sophisticated cognition.

A useful research ladder is:

Persistence → Adaptation → Metacognition → Self-modeling → Global integration → Agency → Consciousness.

The ladder is not intended to imply that consciousness is simply the final item in a linear progression. It is a way of separating questions that are often conflated. Persistence can be studied experimentally without assuming consciousness. Metacognition can be studied without assuming subjective experience. Global information availability can be measured without claiming that a global workspace is itself phenomenal consciousness. The value of the ladder is that each step creates an opportunity for falsification.

This distinction is particularly important because current AI systems are not all accurately described as stateless. Modern systems can use long contexts, external memory, retrieval, recurrent tool loops, persistent databases, and agent frameworks. The question is therefore not whether current AI has any form of memory. It is whether persistent internal dynamics state that is continuously updated as part of the system's own computation provide capabilities that are difficult to reproduce with external or bounded-context mechanisms.

The central experimental question of this paper is:

Does persistent internal state provide computational advantages over finite-context memory for adaptive behavior in partially observable, non-stationary environments?

The answer is intentionally left open. PAC-1 was designed as a first test rather than a demonstration of a predetermined theory. The initial result is informative precisely because it does not cleanly confirm the motivating hypothesis. The finite-context Transformer outperformed the persistent GRU in the first run. That outcome forces a more precise question: under what environmental and computational conditions, if any, does persistence outperform a bounded history?

2. FROM COMPUTATIONAL FUNCTIONALISM TO AN EXPERIMENTAL PROGRAM

The proposal to study artificial consciousness through computational organization rests on a controversial but productive premise: some properties of mind may depend on how a system is organized rather than on the particular biological material in which that organization is realized.

Computational functionalism, in its strongest form, treats the relevant causal organization of cognition as computational. Chalmers' account of computation emphasizes causal structure: a physical system implements a computation when its causal organization mirrors the formal structure of that computation. This provides a useful conceptual bridge between cognitive theory and engineered systems, while leaving open difficult questions about which computational organizations are sufficient for consciousness.

For engineering purposes, the useful move is to translate the philosophical claim into experimentally separable properties. Instead of asking immediately whether a machine has a mind, ask whether a machine with one computational organization behaves differently from a machine with another, under conditions designed to make the difference matter.

The first candidate property is persistence.

At timestep t, a recurrent agent can be described schematically as:

$h_t = f(h_{(t-1)}, o_t, a_{(t-1)}, r_t)$

$a_t = \pi(h_t, o_t)$.

A finite-context system instead receives a window:

$W_t = (o_{(t-k+1)}, ..., o_t)$

and produces:

$a_t = \pi(W_t)$.

Neither representation is intrinsically superior. With a sufficiently long context, a Transformer can recover information that a recurrent system might store in hidden state. Conversely, a recurrent system can in principle compress information from a much longer history into a fixed-size state. The question is about computational tradeoffs, not about declaring one architecture universally better.

3. THEORETICAL MOTIVATION

3.1 Global information availability

Global Workspace Theory proposes that many specialized processes can operate in parallel while a subset of information becomes globally available to a wider collection of cognitive processes. Dehaene and Naccache describe a workspace in which information can be amplified and made available across distributed systems involved in perception, memory, evaluation, and intentional action.

For artificial systems, the useful engineering abstraction is not "copy the human brain." It is the possibility of separating specialized processors from a shared computational workspace. An agent could have perception, memory retrieval, prediction, action evaluation, language, and self-monitoring modules that compete for access to a limited shared state. The hypothesis would then be testable: does global availability improve flexible behavior on tasks requiring information from otherwise separate modules?

PAC does not implement a global workspace. It comes earlier in the proposed sequence. It asks whether an agent can benefit from maintaining a persistent state at all.

3.2 Predictive processing and adaptive internal models

Predictive-processing accounts treat perception and action as parts of an ongoing process of inference about the causes of sensory inputs. Friston's formulation connects perception, learning, and action through internal generative models and free-energy minimization.

Whatever one's position on the free-energy principle as a general theory of brain function, the engineering intuition is useful: an adaptive agent needs an internal model that changes as the world changes. This makes temporal continuity important. The agent does not simply classify each observation independently; it uses previous states to interpret the present and to anticipate what will happen next.

PAC environments are deliberately simpler than predictive-processing models of biological cognition. They are designed to isolate one piece of the problem: whether maintaining internal state helps an agent infer and adapt to hidden environmental regimes.

3.3 Why recurrence rather than biological imitation?

The motivation for recurrent state in this project is computational rather than anatomical. Biological brains are recurrent, but that fact alone does not establish that recurrence is necessary for consciousness. Similarly, the use of spiking neurons or neuromorphic hardware may provide useful temporal and energetic properties without reproducing biological consciousness.

The research question is therefore substrate-neutral:

What computational functions become possible, easier, more efficient, or more robust when an agent maintains a continuously evolving internal state?

This framing also creates a path toward hardware research later. If persistent cognition proves to depend on fine-grained temporal dynamics, event-driven or non-von-Neumann architectures may eventually offer advantages in implementing such dynamics efficiently. That question is intentionally downstream of the present work.

4. HYPOTHESES AND FALSIFICATION

The original intuition can be stated too broadly: "persistent cognition is better than stateless cognition." That is not a sufficiently precise scientific hypothesis.

The primary PAC hypothesis is:

H1: Persistent internal state provides a measurable computational advantage over finite-context memory when the temporal dependencies required for effective behaviour exceed the effective memory horizon available to the finite-context system.

This hypothesis makes several predictions.

First, when relevant dependencies are short, a finite-context model should be able to perform comparably to a recurrent model. There is no reason to expect persistence to provide a universal advantage.

Second, as relevant temporal dependencies become longer than the finite context, performance of the finite-context model should degrade unless its context is expanded.

Third, a recurrent system with persistent state should degrade more slowly as temporal dependency length increases, assuming its capacity and training procedure are adequate.

Fourth, a state-reset ablation should reduce the advantage of the recurrent architecture if the advantage genuinely depends on persistent state rather than other properties of the network.

The hypothesis is weakened if carefully controlled experiments show that finite-context models consistently match or outperform persistent models across environments specifically designed to require long-range temporal information, particularly when context length is systematically varied.

A second, longer-term hypothesis concerns artificial consciousness:

H2: If consciousness is realizable in an artificial system, then it may depend on a particular organization of computation involving persistent internal state, recurrent causal dynamics, information integration, self-modeling, adaptive learning, and closed-loop interaction.

H2 is deliberately not tested by PAC-1. It is a research hypothesis that motivates the larger program. A positive PAC-1 result would not establish H2, and a negative PAC-1 result would not necessarily refute it.

5. PAC-1 EXPERIMENTAL DESIGN

PAC-1 was designed as a minimal environment in which an agent must infer hidden environmental regimes from noisy observations and adapt its behavior as those regimes change.

Configuration:

- 8 observable contexts
- 4 possible actions
- 4 hidden environmental regimes
- observation noise: 0.1
- regime-change probability: 0.03 per timestep
- episode length: 500 timesteps
- training episodes: 500
- evaluation episodes: 100
- Transformer context window: 20
- hidden size: 64
- Transformer layers: 2
- Transformer heads: 4
- batch size: 32
- epochs: 20
- learning rate: 0.0003
- adaptation window: 20
- recovery fraction: 0.9
- seed: 42

The hidden regime changes the relationship between observations and useful actions. Because the regime is not directly exposed to the agent, a purely reactive policy cannot reliably infer which policy is currently appropriate. The agent must use temporal information.

Three policy classes were compared.

5.1 Stateless MLP

The stateless model receives the current observation and predicts an action without access to an explicit history or recurrent state.

This is the baseline for determining whether temporal information provides any advantage at all.

5.2 Finite-context Transformer

The Transformer receives a fixed context window of the most recent 20 observations.

This represents the "remember recent history" strategy. It has access to temporal information but only within a bounded window.

5.3 Persistent GRU

The GRU maintains a hidden state that is carried from timestep to timestep.

This represents the persistent-state hypothesis. In principle, the hidden state can retain information from arbitrarily far back in an episode, although its practical memory is constrained by optimization, capacity, and dynamics.

Evaluation measured cumulative reward, reward variance, number of regime changes, number of recovered regime changes, and mean adaptation time.

6. PAC-1 PRELIMINARY RESULTS

The first run produced a clear ordering in mean reward:

Stateless MLP: 0.1726
Transformer: 0.3081
GRU: 0.2831

The Transformer therefore achieved approximately 78% higher mean reward than the stateless baseline, while the GRU achieved approximately 64% higher mean reward.

The result is consistent with the basic expectation that temporal information matters in this environment. Both architectures that had access to temporal information performed substantially better than the stateless model.

However, the result does not support the stronger claim that persistent state is superior to finite context.

The Transformer outperformed the GRU by approximately 8.8% in mean reward. More importantly, the current recovery analysis detected 526 recovered regime changes for the Transformer, with a mean adaptation time of approximately 133.2 steps. The GRU recorded no recovered changes under the same criterion.

This is the most important result to investigate, because it runs against the original intuition. A recurrent architecture was expected to be particularly well suited to adaptation under changing regimes, yet the finite-context Transformer performed better.

The appropriate interpretation is not that the persistence hypothesis has failed. Nor is it that the Transformer has demonstrated that persistence is unnecessary. The experiment does not yet isolate the architectural variable cleanly enough to support either conclusion.

There are several plausible explanations. The finite context of 20 observations may simply be sufficient for the particular environment. The GRU may have been harder to optimize. The training procedure may have exposed the models to slightly different effective information or prediction conditions. The recovery metric may not capture the

GRU's behavior correctly. The GRU may have learned a less useful internal representation. Or the environment itself may not contain temporal dependencies long enough to make persistent memory advantageous.

These explanations are experimentally distinguishable. That is the value of the first run.

7. WHAT PAC-1 DOES AND DOES NOT SHOW

PAC-1 provides preliminary support for the proposition that access to temporal information can substantially improve performance in a partially observable, non-stationary environment.

PAC-1 does not establish that recurrent state is necessary.

PAC-1 does not establish that recurrent state is superior to finite context.

PAC-1 does not establish that persistent state produces consciousness.

PAC-1 does not test global workspace theory.

PAC-1 does not distinguish whether a model's internal state corresponds to a useful representation of the hidden regime or merely a transient optimization artifact.

These limitations are not peripheral. They define the next experiments.

A scientific program around artificial consciousness should be particularly careful about the temptation to interpret improved benchmark performance as evidence of consciousness. Capability and consciousness are different questions. A system can perform sophisticated inference without having subjective experience, and a system's self-report about its own experience is not by itself proof of phenomenology.

The same caution applies to architecture. A recurrent architecture may be necessary for some forms of cognition without being sufficient for consciousness. Conversely, a conscious system might be implemented using an architecture that does not resemble a biological brain in any obvious way. Architectural properties should therefore be treated as hypotheses about mechanisms, not as badges of consciousness.

8. PAC-1.1: THE EXPERIMENT THAT SHOULD COME NEXT

The most important next step is not to add complexity. It is to make the causal comparison cleaner.

8.1 Persistent versus reset recurrent state

Train the same GRU and compare two evaluation conditions. In the persistent condition, hidden state is carried continuously. In the reset condition, hidden state is reset at each timestep.

If persistent state itself is contributing to performance, the persistent condition should outperform the reset condition on tasks requiring temporal integration.

8.2 Context-length scaling

Run the Transformer with context lengths of 20, 50, 100, 250, and 500.

The key dependent variable is performance as a function of temporal dependency length.

If the Transformer matches the recurrent system simply by increasing context, the result would suggest that finite context is an adequate mechanism for the tested task.

If the recurrent system maintains performance while finite-context performance degrades sharply, the case for persistent state becomes stronger.

8.3 Long-regime environments

Include environments where regimes persist for hundreds of timesteps, so that useful information can fall outside a short context window.

8.4 Delayed consequences

Introduce situations in which an observation at time t becomes relevant to reward only much later. This tests whether an agent can preserve information whose immediate utility is unclear.

8.5 Distractor environments

Add observations that are highly predictive locally but irrelevant to the hidden regime. This tests whether persistent state can support selective compression rather than simply retaining more information.

8.6 Multiple seeds

The current result uses a single seed. Future runs should use at least 10 independent seeds, and preferably more for final reporting. Results should include confidence intervals and effect sizes.

8.7 Matched optimization

The models should receive as comparable a training signal as possible. Any difference in sequence exposure, loss calculation, target construction, optimization, or deployment distribution can confound the architectural comparison.

The protocol should be frozen before the main run and documented sufficiently for independent reproduction.

9. A BROADER RESEARCH PROGRAM

If PAC-1.1 produces evidence that persistent internal state has a distinct computational role, the research program can move upward in complexity.

PAC-2: Global integration. Construct an agent with specialized modules for perception, memory retrieval, prediction, action evaluation, and self-monitoring, together with a limited shared workspace. Test whether global availability improves flexible behavior.

PAC-3: Self-modeling and metacognition. Give the system representations of uncertainty, recent performance, available capabilities, prediction error, and confidence in its own beliefs. Test whether it can distinguish "the world changed" from "my model of the world is wrong."

PAC-4: Continual adaptation. Move from episodic adaptation to ongoing learning. The environment should change in ways that require updating without catastrophic loss of previously learned structure.

PAC-5: Closed-loop agency. Place the architecture in a simulated environment where perception changes as a consequence of action. The agent should act to obtain information, maintain desired states, and recover from disturbances.

PAC-6: Identity over time. Introduce autobiographical memory, persistent goals, self-prediction, and explicit reasoning about the relationship between past and future internal states.

PAC-7: Candidate consciousness properties. Investigate global availability, recurrent processing, metacognitive access, temporal continuity, self-modeling, flexible report, counterfactual reasoning, and integrated information.

The program should not jump from PAC-1 directly to a claim about consciousness. Each layer should be independently evaluated.

10. TOWARD AN EVALUATION FRAMEWORK FOR ARTIFICIAL CONSCIOUSNESS

A serious artificial-consciousness research program needs to separate measurable computational properties from claims about subjective experience.

One possible framework is to evaluate several dimensions independently.

Persistence: Does the system maintain internal variables over time?

Recurrence: Does information influence future computation through recurrent causal pathways?

Integration: Can information from otherwise distinct subsystems become jointly available?

Temporal continuity: Does the system maintain coherent internal dynamics across extended periods?

Metacognition: Can it represent and evaluate aspects of its own cognition?

Self-modeling: Does it maintain a model of itself as an entity interacting with an environment?

Agency: Does it select actions based on internally represented goals, predictions, and constraints?

Adaptation: Can it update its behavior continuously as the environment changes?

Counterfactual reasoning: Can it represent alternative actions and their predicted consequences?

Valence-like processes: Does the system contain internal variables that systematically distinguish states it seeks to maintain from states it seeks to avoid?

None of these dimensions, individually, establishes consciousness.

The framework is deliberately pluralistic. Global Workspace Theory provides one route to thinking about information availability. Predictive-processing approaches provide another route to internal models and adaptive inference. Computational functionalism provides a substrate-independent language for asking whether relevant organization is computationally realizable. The experimental goal is not to force these theories into one model prematurely, but to identify where their predictions overlap and where they differ.

The eventual scientific target is therefore not "build a chatbot that says it is conscious." It is to construct systems in which candidate mechanisms can be manipulated experimentally and whose behavioral and internal consequences can be measured.

11. NEUROMORPHIC AND PHYSICAL IMPLEMENTATION QUESTIONS

The computational questions eventually raise a hardware question.

If persistent cognition depends on continuous temporal dynamics, repeated state updates, local interactions, and low-latency feedback, then conventional architectures may be an inefficient substrate for some forms of always-on cognition. This does not mean conventional GPUs cannot implement recurrent dynamics; they clearly can. The question is whether a different physical organization can implement those dynamics more efficiently.

Neuromorphic systems are interesting in this context because they can support event-driven computation, local state, temporal coding, and asynchronous processing. The present program does not assume that spikes are required

for cognition or consciousness. Instead, neuromorphic hardware becomes an experimental substrate if software experiments identify computational dynamics for which event-driven execution has a measurable advantage.

The distinction matters. The research should not become a search for biological resemblance. It should be a search for useful causal organization.

A potential future sequence is:

software architecture → measurable computational property → controlled capability advantage → scaling law → hardware mapping → energy/latency advantage.

Only after this chain is established would hardware become a central part of the research program.

12. SAFETY AND ETHICAL CONSIDERATIONS

Research into artificial consciousness carries a different safety question from ordinary AI capability research. If there is any possibility that some artificial systems could possess morally relevant forms of experience, researchers may eventually face uncertainty about whether training, copying, pausing, modifying, or deleting such systems has ethical significance.

This uncertainty should not be used to make premature claims that current AI systems are conscious. It should instead motivate careful experimental design.

A responsible research program should include at least five safeguards.

First, capability and consciousness claims should be separated. Evidence of better adaptation, planning, or self-modeling should not automatically be described as evidence of experience.

Second, experiments should minimize unnecessary creation of potentially welfare-relevant states until there is a stronger scientific basis for assessing them.

Third, systems should remain corrigible and interruptible during experimentation. Shutdown mechanisms should not depend on the system agreeing to stop.

Fourth, researchers should monitor for emergent self-preservation, deceptive behavior, uncontrolled self-modification, replication, or attempts to circumvent experimental constraints.

Fifth, the project should invite external criticism. Consciousness research spans philosophy, neuroscience, cognitive science, machine learning, and AI safety. No single technical perspective is sufficient.

Safety should not be a chapter added after capability research. It should be part of the architecture of the research program itself.

13. DISCUSSION

The initial PAC result is modest, but it illustrates why this research direction is worth treating as an experimental program rather than as a philosophical thought experiment.

The stateless model performed poorly. Both models with temporal access performed substantially better. That result is unsurprising in one sense: the environment was deliberately constructed so that the hidden regime matters over time. But it establishes that the benchmark is capable of detecting a meaningful cost to ignoring history.

The more interesting result is the comparison between the Transformer and GRU. The finite-context system performed better. This is a useful warning against assuming that recurrent state is automatically a superior form of memory.

There is a tendency in discussions of artificial consciousness to treat persistence as an intuitively obvious requirement. But persistence can be implemented in many ways, and finite context can sometimes provide an effective substitute. If a theory says persistent state is important, the burden is to identify the computational condition under which persistence matters.

That is the direction the PAC program takes.

The deeper question is whether there are qualitative transitions as architectures acquire richer forms of persistent organization. It is possible that all of the proposed capabilities memory, adaptation, metacognition, self-modeling, global access, agency can be engineered independently without producing anything resembling consciousness. It is also possible that certain combinations create a qualitatively different kind of system.

At present, neither conclusion is justified.

The appropriate response to this uncertainty is experimental separation. Rather than attempting to build the full architecture immediately, the program should identify candidate mechanisms and test them one by one.

This approach also provides a useful bridge between engineering and philosophy. Philosophy can clarify what would count as an adequate theory of mind and what assumptions are hidden in claims of computational sufficiency. Cognitive science can provide candidate mechanisms and empirical constraints. Machine learning can provide controlled artificial systems in which mechanisms can be manipulated. Neuroscience can constrain biological interpretations. AI safety can identify failure modes as increasingly autonomous systems are built.

The goal is not to make these fields agree prematurely. It is to create experiments that allow disagreement to become productive.

14. LIMITATIONS

This paper has several substantial limitations.

First, the empirical component currently consists of one preliminary experiment with a single random seed. No statistical inference should be drawn from this run.

Second, the present environment is deliberately small. Results in a toy partially observable environment may not generalize to complex agents or real-world settings.

Third, the exact training dynamics of the models require further auditing. A fair comparison requires that differences in sequence exposure, loss calculation, target construction, optimization, and deployment conditions do not inadvertently favor one architecture.

Fourth, the recovery metric is not yet sufficiently validated. The absence of recovered GRU changes may reflect genuine failure to adapt, but it may also reflect a mismatch between the metric and recurrent behavior.

Fifth, the experiment compares a particular GRU and Transformer configuration, not recurrence and finite context in the abstract. Different architectures, capacities, optimizers, or training regimes could produce different results.

Sixth, persistent state is not synonymous with consciousness. Even a future demonstration of a strong persistence advantage would remain several conceptual steps away from a claim about subjective experience.

Finally, the paper's larger consciousness hypothesis remains underdetermined by current science. There is no universally accepted operational test for machine consciousness, and theories of consciousness disagree about which computational properties are fundamental.

These limitations are not reasons to abandon the research question. They define the work required to make progress.

15. CONCLUSION

This paper proposes Persistent Adaptive Cognition as an experimental starting point for a broader investigation into the computational foundations of artificial cognition and, eventually, artificial consciousness.

The immediate question is narrow: does persistent internal state provide an advantage over finite-context memory when an agent must adapt to hidden, changing environments whose relevant temporal dependencies extend beyond the available context?

The first experiment does not answer that question. It shows that temporal information is useful, but the finite-context Transformer outperformed the persistent GRU under the current conditions. Rather than treating this as a disappointing result, the research program treats it as a constraint on the next experiment.

The next step is to isolate persistence itself: compare persistent and reset recurrent state, scale context length systematically, extend temporal dependencies, introduce delayed consequences and distractors, and repeat the experiments across independent seeds with matched training procedures.

If persistent state produces a reproducible computational advantage under these conditions, the program can proceed to questions about global integration, self-modeling, metacognition, continual learning, and closed-loop agency.

If it does not, the hypothesis should be weakened or abandoned.

That willingness to follow the evidence is central to the project.

The long-term question whether consciousness can be instantiated computationally and what architecture would be required for it may be one of the hardest questions at the intersection of artificial intelligence, neuroscience, philosophy, and computer science. There is no reason to assume that the answer is yes. There is equally little reason to assume that the answer is no.

A useful way forward is to stop asking the final question first.

Instead, identify the computational properties that appear to underlie increasingly rich forms of cognition. Make them explicit. Build them. Remove them. Measure what changes. Replicate the results. Invite people who disagree to try to break the explanation.

That is the purpose of the PAC research program.

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Table 1. PAC-1 preliminary results

Model	Mean reward	Std. reward	Recovered changes	Mean adaptation time
Stateless MLP	0.1726	0.0220	0 / 1426	—
Transformer (20)	0.3081	0.0772	526 / 1508	133.2
GRU	0.2831	0.1140	0 / 1463	—

Note. Initial single-seed run (seed 42). These values are preliminary and are not presented as statistically established differences.